

Properties of the Matching Function

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m : matching function

$M = m(S, B)$

$\# \text{ sellers}$ (points to S)

$\# \text{ buyers}$ (points to B)

$\#$ of matches/trades in a given period

Example : labor market

- $M = \#$ of hires
- $S = \#$ unemployed workers
- $B = \#$ vacant jobs

Assumptions about m

- $\frac{\partial m}{\partial S} > 0$, $\frac{\partial m}{\partial B} > 0$
- $m(0, B) = m(S, 0) = 0$
- $\frac{\partial^2 m}{\partial S^2} < 0$, $\frac{\partial^2 m}{\partial B^2} < 0$
- m has constant returns to scale

$$m(\lambda S, \lambda B) = \lambda \cdot m(S, B)$$

- discrete time $M = \#$ trade within time period
- $$m(S, B) \leq \min(S, B)$$

- continuous time $M =$ flow of trades (no restrictions)